Isaac K Cross

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EDUCATION

Northwestern University — Evanston, IL
B.S., Physics, Advanced Physics. 3.96/4.0 — 2026
B.S., Applied Mathematics. 3.98/4.0 — 2026

RESEARCH EXPERIENCE

The Odom Group, The Center for Fundamental Physics, Northwestern University
Undergraduate Researcher, Winter 2023 – Present
Advisor: Prof. Brian C. Odom
Doped Cryogenic Crystals for Quantum Sensing and Fundamental Physics Project

Dual-Species Rubidium-Argon Beam Source for Controlled Cryogenic Crystal Growth
Winter 2023 – Present

- Designed novel, highly collimated, tunable, dual-species Rb-Ar beam source with microcapillary nozzle for the controlled growth of doped cryogenic rare-gas crystals.
- Characterized half-angle divergence of $(\alpha)_{1/e} = 0.74 \pm 0.1$ degrees.
- Worked with PhD student on crystal growth chamber development: compatibility for cryogenic, x-ray diffraction, and optical access set beam source constraints.
- Assembled Rb absorption spectroscopy setup from scratch for characterization: recommissioned ECDL, implemented frequency-modulation control across the ^{85}Rb D_2 F=3→F′=4 transition, designed and aligned complete optics chain (Faraday isolation, mode shaper, collimation, reference cell, PM single-mode fiber coupling).
- Assembled dedicated UHV chamber for characterization: achieved 7.5×10^{-10} torr with turbo, clean, bake, and helium leak test.
- Developed Ar gas handling system introducing Ar to beam source and UHV chamber (Leak checked < 10^{-9} mbarr L/s), allowing tunable Ar flux of ~10^{15} - 10^{18} Ar atoms/s.
- Developed custom mica-nichrome heaters and temperature sensors to control Rb flux and prevent nozzle clogging: achieved ~ 400 °C, week-long operation during UHV bake.
- Designed AC coupler and high-amplifying circuit for spectroscopic data acquisition.
- Prepared Standard Operating Procedures (SOP) for safe operation of ECDL laser and loading of Rb into beam source.

Spin-Spin Interaction Induced Inhomogeneous Broadening Monte Carlo Simulations
Fall 2025 – Present

- Implemented Monte Carlo simulations of dipole-dipole spin interactions between dopant atoms in dilute spin ensembles to predict inhomogeneous broadening.
- Extracted effective coherence times (T2*) from simulated and published linewidths at different dopant densities, enabling analysis of relative contributions to inhomogeneous broadening.
- Identified ideal dopant densities and target coherence times for improved quantum-projection-noise-limited sensitivity.

Kovachy Research Group, Center for Fundamental Physics, Northwestern University
Undergraduate Research Intern, Spring 2023 – Fall 2023
Advisor: Prof. Timothy L. Kovachy

High-Temperature Strontium Vapor Cell
Spring 2023 – Fall 2023

- Designed and machined stainless-steel Sr reservoir with thermal isolation stages for a high-temperature vapor cell used to frequency-lock lasers in a Sr MOT atom-interferometry experiment.
- Implemented custom Nichrome-mica heaters reaching ~580 °C (above 500 °C required for sufficient optical density) and tested in a UHV outgassing chamber with RGA.
- Designed absorption spectroscopy optical layout for Sr vapor cell.

AWARDS AND HONORS

- Junior Career Award in Physics and Astronomy for one Junior with exceptional performance and potential in physics or astronomy, Northwestern, 2025
- Fletcher Undergraduate Research Grant Prize semi-finalist for outstanding research, Northwestern, 2025
- Summer Undergraduate Research Grant (SURG), Northwestern, 2025
- Tau Beta Pi Engineering Honor Society, Inducted 2024
- Weinberg College Summer Research Grant, Northwestern, 2023
- Award for Excellence in Mathematics by a First-Year Student, Northwestern, 2023
- Dean's List High Honors, Northwestern, Spring 2024-Spring 2025
- Edwin Munro Goodman Scholarship, Northwestern, 2025-2026
- Edwin Quackenbush Foundation Scholarship, Northwestern,2023-2025
- J.G. Nolan Scholarship, Northwestern, 2023-2024

PRESENTATIONS AND POSTERS

Towards Doped Cryogenic Rare Gas Crystals for Precision Measurement, Poster, Midwest Cold Atoms Workshop 2025 (November 2025)

Modeling Inhomogeneous Broadening From Spin-Spin Interactions in Matrix Isolated Spin Ensembles, Poster and Term Paper, Northwestern Applied Mathematics 421 Modeling Poster Presentation (December 2025)

[Title TBD], Student Talk, Center for Fundamental Physics Seminar (Winter 2026)

SUMMER SCHOOLS AND WORKSHOPS

Johns Hopkins University Summer School — Baltimore, MD
Two-week Quantum Sensing and Precision Measurement Summer School. — August 2025

Midwest Cold Atoms Workshop (MCAW 2025) — Urbana-Champaign, IL
Two-day cold-atom and quantum science workshop. — November 2025

SELECTED ACADEMIC PROJECTS

Alkali Doped Cryogenic Rare Gas Crystals as a Possible Platform for Solid State Atomic Clocks, Term Paper (Advanced Topics in AMO Physics, Fall 2025), Derived Allan deviation and sensitivity limits for matrix-isolated alkali microwave clocks.

Electroweak Unification, Term Paper and Presentation (Particle Physics, Fall 2024), Derived SU(2)xU(1) gauge formalism unifying weak and electromagnetic interactions; explained spontaneous symmetry breaking and Higgs mechanism.

To The Development of the Theoretical Model and Understanding of Rotating Black Holes, Term Paper (Differential Geometry and Tensor Calculus, Fall 2024), Derived Schwarzschild and Kerr metrics from Einstein's field equations.

Pulsed NMR and 1D Imaging Experiment, Lab and Report (Advanced Physics Lab, Winter 2025), measured T1 and T2 with spin-echo and performed basic 1D spin-density imaging.

Gamma-Gamma Coincidence Measurement Experiment (Na-20, Co-60), Lab and Report (Advanced Physics Lab, Winter 2025), measured γ-γ coincidences and angular correlation.

Second Harmonic Generation Experiment, Lab and Report (Optics Lab, Fall 2024), Characterized SHG power dependence on input power, temperature, and wavelength.

Fourier Optics Experiment, Lab and Report (Optics Lab, Fall 2024), Demonstrated Fourier-transforming properties of optics and studied filtering in the Fourier plane. Tested custom resin 3-D printed optics.

Optimal Control Multi-Joint Robotic Arm, Modeling Project Final (Modeling and Computation in Science and Engineering, Winter 2025), Developed numerical method to find minimum-energy trajectories and torques for coupled nonlinear mechanical arm system.

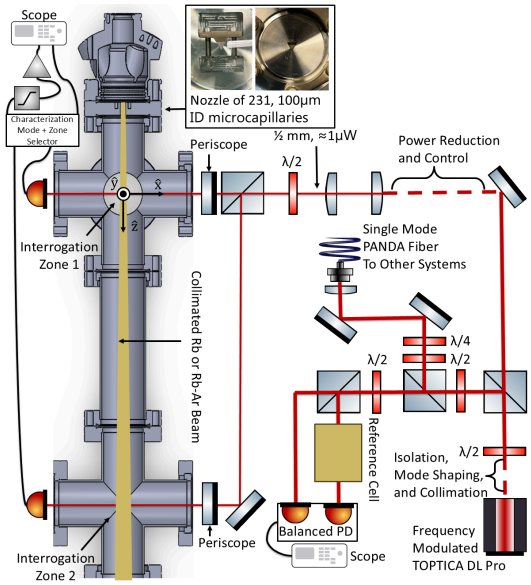
ECG Heart Monitor, Engineering Project Final (Intro. Electrical Engineering, Fall 2023), designed and built a three-stage instrumentation amplifier to record and analyze my ECG signal.

TECHNICAL SKILLS AND EXPERIENCE

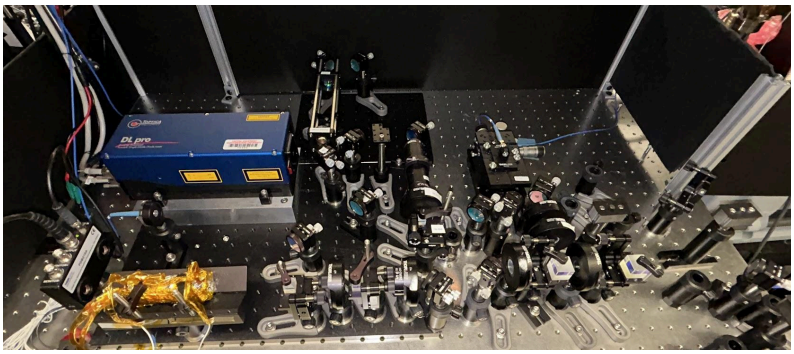
Experimental: Atomic oven design, UHV chamber design and assembly, absorption and Doppler-free spectroscopy, ECDL operation, optical breadboard layout/alignment, alkali metals handling, gas-handling systems, Helium leak testing, lock-in detection, pulsed NMR, writing standard operating procedures, machine-shop work (lathe, mill), 3D printing (FDM and SLA), analog electronics for measurement, high-temperature heater design, mentoring new students.
Computational & Programming: Python (NumPy/SciPy/matplotlib), Mathematica, MATLAB, Arduino, LaTeX, CAD (SolidWorks, AutoCAD), Blender for scientific illustration, Monte Carlo methods, numerical solutions of ODEs and PDEs, optimization, data analysis, curve fitting, uncertainty and error analysis.
Certifications: Machine Shop Training, Laser Safety Certification, Laboratory Safety Certification, Hazardous Chemical Waste Management Certification, Personal Protective Equipment Certification.
Relevant Coursework: Physics – Advanced Topics in AMO Physics, Advanced Quantum Electronics, Advanced Physics Laboratory, Optics Laboratory, Quantum Mechanics, Particle and Nuclear Physics, Advanced Electricity & Magnetism, Statistical Mechanics; *Mathematics* – Modeling and Computation in Science & Engineering, Models in Applied Mathematics, Differential Geometry and Tensor Calculus, Ordinary and Partial Differential Equations, Complex Analysis, Fourier Analysis and Boundary Value Problems, Applied Nonlinear Dynamical Systems, Linear and Applied Linear Algebra, Full Calculus Sequence.

APPARATUS DIAGRAM AND IMAGES

Simplified diagram of dual-species beam characterization chamber and spectroscopy set up:



Rubidium spectroscopy setup for dual-species beam characterization:



Dual-species characterization UHV chamber and beam source:

